Scalable Robot Co-Design via Performance-Gated Hybrid Optimization

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***Abstract*—We present Performance-Gated Hybrid Co-Design (PGHC), a scalable framework for simultaneous optimization of robot morphology and control policies. Existing co-design approaches face fundamental tradeoffs: evolutionary methods achieve generality but suffer from exponential sample complexity, while differentiable physics promises gradient-based optimization but fails in practice when morphology parameters interact with contact dynamics. PGHC bridges this gap through a theoretically motivated combination of three elements: (1) deep RL for policy optimization, (2) the Envelope Theorem to justify freezing a converged policy during morphology updates, and (3) finite-difference morphology gradients enabling first-order gradient descent on design parameters—replacing the population-based search used by all prior co-design methods. Dynamic stability gating and adaptive trust regions enforce the conditions under which gradient decoupling remains valid. We demonstrate PGHC on full-body humanoid morphology optimization, achieving 12–18% improvements in energy efficiency across diverse locomotion tasks while maintaining motion quality. To our knowledge, this is the first application of the Envelope Theorem to RL-based robot co-design, the first use of gradient descent (rather than evolutionary search) for the morphology outer loop, and the first demonstration of full humanoid co-design with deep RL policies. We additionally document a negative result: differentiable physics via BPTT fails to produce useful morphology gradients through contact, explaining why the field has historically defaulted to gradient-free methods.**

I. Introduction

Co-design methods simultaneously optimize robot hardware and control policies to ensure task-specific performance. However, scaling these approaches to high-dimensional morphology spaces with complex learned controllers remains an open challenge. Recent surveys [**?**] catalog over 100 co-design papers, yet none demonstrate full humanoid co-design with deep RL, use the Envelope Theorem for gradient decoupling, or employ first-order gradient descent on morphology parameters. Liu et al. [**?**] explicitly identify humanoid co-design as "an important but largely unexplored challenge."

A. The Scalability Challenge

Existing co-design approaches face a fundamental tradeoff between generality and scalability:

Evolutionary methods [**?**], [**?**], [**?**], [**?**], [**?**], [**?**]: Gradient-free approaches (CMA-ES, genetic algorithms, evolution strategies) treat design evaluation as a black box, achieving broad applicability. However, they require full policy retraining for *every* candidate design, with sample complexity scaling poorly in design dimensionality. While successful for low-DOF subsystems (quadruped legs [**?**], grippers [**?**], voxel creatures [**?**]), the cost of repeated retraining makes them expensive for high-dimensional morphology spaces. Notably, *every* existing co-design method that pairs RL with morphology optimization—including DERL [**?**], RoboCraft [**?**], and TAME [**?**]—uses a zeroth-order outer loop.

Differentiable physics methods [**?**], [**?**], [**?**], [**?**]: These approaches promise efficient gradient-based optimization by backpropagating through the physics simulation. However, in practice, differentiating through contact dynamics with respect to morphology parameters is problematic: constraint solvers (e.g., in MuJoCo/MJX) do not reliably differentiate with respect to body parameters, producing uninformative or numerically unstable gradients. Prior work using analytical gradients [**?**], [**?**] has been limited to trajectory optimization on simple systems, not deep RL policies. We confirm this limitation empirically (Section III).

The missing middle ground: No prior work combines all of: (1) a bilevel structure with RL inner loop and morphology outer loop, (2) the Envelope Theorem to justify freezing the policy, (3) first-order gradient descent on morphology via finite differences, and (4) application to a full humanoid. PGHC fills this gap.

B. Our Approach

We propose PGHC, a hybrid framework whose core insight is that **the Envelope Theorem, applied to a converged RL policy, eliminates the need for both costly implicit differentiation and population-based search**. When the policy is locally optimal, the total derivative of performance with respect to morphology reduces to the direct (partial) derivative—the implicit term vanishes. This justifies a simple but powerful strategy: train the RL policy to convergence, freeze it, estimate morphology gradients via finite differences, and update the design via gradient descent.

This observation, while grounded in classical optimization theory, has not been applied to RL-based robot co-design. The practical gap between the textbook theorem and a working algorithm on a full humanoid is bridged by two mechanisms—**dynamic stability gating** (ensuring the policy is actually converged before morphology updates) and **adaptive trust regions** (ensuring morphology updates remain within the basin of attraction of the current policy).

Key contributions:

1) **Envelope Theorem for RL co-design**: First application of the Envelope Theorem to justify gradient decoupling in RL-based robot co-design, enabling a frozen-policy morphology optimization strategy that no prior work has demonstrated.

2) **First-order morphology gradient descent**: First co-design framework to use gradient descent (via finite differences) rather than population-based search for the morphology outer loop, yielding fundamentally different convergence properties and lower per-step cost.
3) **Full humanoid co-design**: First demonstration of co-design optimization on a full humanoid (30+ DOF) with deep RL policies across diverse locomotion behaviors.
4) **BPTT negative result**: Empirical documentation of why differentiable physics via BPTT fails for morphology optimization through contact, explaining the field's historical reliance on gradient-free methods.
5) **Comprehensive evaluation**: Sample efficiency comparisons, ablation studies validating each framework component, and empirical analysis of gradient approximation quality.

Evaluation focus: While our framework is agnostic to morphology parameters and task domains, we focus our primary evaluation on humanoid locomotion for two reasons: (1) it represents one of the most challenging co-design scenarios (contact-rich dynamics, high DOF, stability constraints), and (2) it enables integration with motion priors (AMP [**?**]) that prevent morphology exploitation while ensuring human-like motion. We demonstrate generality through secondary experiments on link length optimization.

II. Methodology

A. Problem Formulation

We formulate co-design as a bilevel optimization problem. Let $\phi \in \mathbb{R}^n$ represent morphology parameters and $\theta \in \mathbb{R}^m$ represent policy parameters. The objective is to minimize a task-specific cost $\mathcal{J}$ (e.g., energy consumption, time-to-goal) while ensuring effective control:

$$\min_{\phi} \mathcal{J}(\phi, \theta^*(\phi)) \quad \text{s.t.} \quad \theta^*(\phi) = \arg\max_{\theta} \mathbb{E}_{\tau}[R(\tau;\phi)] \tag{1}$$

where $R(\tau;\phi)$ is the cumulative reward for trajectory τ under morphology ϕ. This formulation is general—applicable to any morphology parameterization and any RL algorithm that learns policy parameters θ.

B. The Bilevel Optimization Challenge

Ideally, optimizing ϕ requires the total derivative:

$$\frac{d\mathcal{J}}{d\phi} = \underbrace{\frac{\partial \mathcal{J}}{\partial \phi}}_{\text{Direct Term}} + \underbrace{\frac{\partial \mathcal{J}}{\partial \theta^*}\frac{\partial \theta^*}{\partial \phi}}_{\text{Implicit Term}} \tag{2}$$

The implicit term $\frac{\partial \theta^*}{\partial \phi}$ requires differentiating through the entire policy optimization trajectory—computing second-order derivatives, storing full training history, and calculating Hessian-vector products. For deep RL policies with millions of parameters trained over millions of timesteps, this is computationally intractable.

The field has addressed this in two ways, both with significant limitations:

Algorithm 1 Performance-Gated Hybrid Co-Design (PGHC)

Input: Initial design ϕ_0, Initial policy θ_0
Input: Task reward function r_{task}, Optional: Motion data $\mathcal{M}$
Input: Learning rates α (policy), β_{init} (design)
Input: Thresholds: Stability δ_{conv}, Trust Region ξ
Input: FD perturbation ϵ_{fd}, FD type (central/forward/SPSA)
Hyperparameters: Window W, Evaluation episodes N_{eval}
Initialize physics environment with ϕ_0
Initialize policy π_θ and RL algorithm (e.g., PPO, SAC, AMP)
$\beta \leftarrow \beta_{init}$
$k \leftarrow 0$
while not converged **do**
  // Phase 1: Performance-Gated Inner Loop (RL)
  repeat
    $\mathcal{D} \leftarrow \text{Rollout}(\pi_\theta, \phi_k, T_{rollout})$
    Compute total reward r_{total} (task + optional style)
    Update Policy $\theta \leftarrow \theta + \alpha \nabla_\theta L^{RL}$
    Update Moving Avg Return $\bar{R}_t$ over window W
    $\Delta_{rel} \leftarrow \frac{|\bar{R}_t - \bar{R}_{t-W}|}{|\bar{R}_{t-W}| + \epsilon}$
  until $\Delta_{rel} < \delta_{conv}$ ▷ Stability Gate
  Let $\theta^* \leftarrow \theta$ ▷ Converged policy (frozen)
  // Phase 2: Finite-Difference Morphology Gradient
  // Envelope Theorem justifies: $\frac{d\mathcal{J}}{d\phi} \approx \frac{\partial \mathcal{J}}{\partial \phi}\big|_{\theta=\theta^*}$
  for each design parameter ϕ_i **do**
    $\phi^+ \leftarrow \phi_k; \quad \phi_i^+ \leftarrow \phi_{k,i} + \epsilon_{fd}$
    $\phi^- \leftarrow \phi_k; \quad \phi_i^- \leftarrow \phi_{k,i} - \epsilon_{fd}$
    $\hat{g}_i \leftarrow \frac{\mathcal{J}(\phi^+, \theta^*) - \mathcal{J}(\phi^-, \theta^*)}{2\epsilon_{fd}}$ ▷ Central FD
  end for
  $\nabla_\phi \mathcal{J} \leftarrow \hat{g}$ ▷ FD gradient estimate
  // Phase 3: Adaptive Trust-Region Design Update
  repeat
    $\phi' \leftarrow \phi_k - \beta \nabla_\phi \mathcal{J}$ ▷ Candidate Design
    $D \leftarrow \mathcal{J}(\phi', \theta^*) - \mathcal{J}(\phi_k, \theta^*)$
    if $|D| > \xi \cdot |\mathcal{J}(\phi_k, \theta^*)|$ **then** ▷ Violation
      $\beta \leftarrow \beta \cdot 0.5$
    else
      $\phi_{k+1} \leftarrow \phi'$
      if $|D|$ is small **then** $\beta \leftarrow \beta \cdot 1.5$
      end if
      Break
    end if
  until accepted
  $k \leftarrow k + 1$
end while
Return optimized design ϕ^* and control policy θ^*

- **Avoid gradients entirely**: Use population-based search (CMA-ES, GA, ES) for the outer loop, treating the entire inner RL training as a black-box evaluation. This is the dominant approach [**?**], [**?**], [**?**], but requires retraining the policy from scratch for every candidate design.
- **Differentiate through physics**: Use BPTT through a differentiable simulator to compute $\frac{\partial \mathcal{J}}{\partial \phi}$ analytically. This has been demonstrated for trajectory optimization on simple systems [**?**], [**?**], but fails for morphology parameters that interact with contact dynamics (Section III).

C. Gradient Decoupling via the Envelope Theorem

We propose a third path. The Envelope Theorem from optimization theory [**?**] provides a principled resolution: at a local optimum θ^*, the gradient of the objective with respect to policy parameters vanishes ($\nabla_\theta \mathcal{J}(\phi, \theta^*) \approx 0$). Consequently, the implicit term drops out:

$$\frac{\partial \mathcal{J}}{\partial \theta^*} \underbrace{\frac{\partial \theta^*}{\partial \phi}}_{\text{implicit}} \approx 0 \quad \text{when } \theta^* \text{ is locally optimal} \tag{3}$$

This yields the key approximation:

$$\frac{d\mathcal{J}}{d\phi} \approx \left.\frac{\partial \mathcal{J}}{\partial \phi}\right|_{\theta=\theta^*} \tag{4}$$

Practical implication: Once the RL policy has converged, we can *freeze* it and estimate morphology gradients by evaluating (not retraining) the frozen policy under perturbed designs. This eliminates both the intractable implicit differentiation *and* the need for population-based search with full retraining per candidate.

Novelty: While the Envelope Theorem is well-known in economics and has been applied in meta-learning, *no prior robot co-design work has invoked it to justify frozen-policy morphology optimization*. The closest conceptual relative is Strgar et al. [**?**], who freeze a pretrained controller for "zero-shot evolution"—but they use evolutionary search, not gradient descent, and operate on voxel-based soft creatures rather than humanoids.

D. Finite-Difference Morphology Gradients

With the frozen policy θ^*, we estimate the morphology gradient via central finite differences:

$$\left.\frac{\partial \mathcal{J}}{\partial \phi_i}\right|_{\theta^*} \approx \frac{\mathcal{J}(\phi + \epsilon_{fd}\mathbf{e}_i, \theta^*) - \mathcal{J}(\phi - \epsilon_{fd}\mathbf{e}_i, \theta^*)}{2\epsilon_{fd}} \tag{5}$$

where $\mathbf{e}_i$ is the i-th standard basis vector and ϵ_{fd} is the perturbation magnitude. Each evaluation involves rolling out the frozen policy θ^* under the perturbed morphology and averaging the objective over N_{eval} episodes to reduce variance.

Why FD instead of BPTT? Differentiable physics (BPTT through the simulator) would in principle provide analytical gradients. However, we found empirically that current differentiable simulators (MJX) fail to produce useful gradients for morphology parameters that affect contact geometry (Section III). Finite differences are agnostic to the simulator's internal structure—they only require the ability to *evaluate* the objective, not differentiate through it.

Why FD instead of evolutionary search? Unlike CMA-ES or genetic algorithms, FD-based gradient descent provides:

- **First-order convergence**: Gradient descent converges at rate $O(1/k)$ vs. the slower rates of zeroth-order methods in the same dimensionality.
- **No retraining**: Each FD evaluation uses the frozen policy—a rollout, not a full training run. For n design parameters, central FD requires $2n$ rollouts per gradient step, while CMA-ES requires $O(\text{population size})$ full retraining runs.
- **Scalability**: For higher-dimensional design spaces, Simultaneous Perturbation Stochastic Approximation (SPSA) [**?**] estimates the gradient with only 2 evaluations regardless of dimensionality.

Cost analysis: For n design parameters with central FD, each outer step requires $2n+1$ rollout evaluations of the frozen policy ($\sim$seconds each). Compare to CMA-ES, where each candidate requires full policy retraining ($\sim$hours each). Even for modest population sizes, the per-step cost difference is orders of magnitude.

E. Performance-Gated Stabilization

The Envelope Theorem approximation is valid when:

1) The policy is locally optimal: $\nabla_\theta \mathcal{J}(\phi, \theta^*) \approx 0$
2) Design updates preserve policy validity: θ^* remains in its basin of attraction

We enforce these conditions through two mechanisms:

1) Stability Gating: We define a performance stability metric based on moving average episode returns:

$$\Delta_{rel} = \frac{|\bar{R}_t - \bar{R}_{t-W}|}{|\bar{R}_{t-W}| + \epsilon} \tag{6}$$

where $\bar{R}_t$ is the average return over the most recent W episodes. Morphology updates trigger *only when* $\Delta_{rel} < \delta_{conv}$, ensuring the policy has converged locally before computing design gradients. We use $W = 100$ episodes and $\delta_{conv} = 0.05$ (5% relative change).

This is critical: without stability gating, the Envelope Theorem approximation is invalid and morphology gradients become unreliable. Fixed-iteration heuristics (e.g., "train policy for N steps") cannot guarantee convergence across different morphologies and tasks.

2) Adaptive Trust Region: For a candidate design ϕ', we compute the performance change under the frozen policy:

$$D = \mathcal{J}(\phi', \theta^*) - \mathcal{J}(\phi_k, \theta^*) \tag{7}$$

The update is accepted only if:

$$|D| \leq \xi \cdot |\mathcal{J}(\phi_k, \theta^*)| \tag{8}$$

If violated, we decay the design step size $\beta \leftarrow 0.5\beta$, constraining morphology changes to regions where the current policy remains effective. We use $\xi = 0.1$ (10% performance degradation threshold). This ensures morphology updates stay

within the basin of attraction of θ^*, maintaining the validity of the frozen-policy evaluation.

Together, these mechanisms create a safe optimization trajectory:

$$\nabla_\phi \mathcal{J} \approx \frac{\partial \mathcal{J}}{\partial \phi}\bigg|_{\theta=\theta^*, \Delta_{rel}<\delta_{conv}} \tag{9}$$

III. BPTT Negative Result

Before arriving at the finite-difference approach, we attempted morphology gradient computation via backpropagation through time (BPTT) in a differentiable physics simulator (MuJoCo MJX). We document this negative result because it explains a gap in the literature: why no prior RL-based co-design work uses analytical morphology gradients through contact.

Setup: We implemented BPTT through MJX rollouts, computing $\frac{\partial \mathcal{J}}{\partial \phi}$ by differentiating through the simulator's forward dynamics with respect to body parameters (link lengths, joint orientations).

Findings:

1) **Contact solver discontinuities**: MJX's constraint solver does not reliably differentiate with respect to body parameters that affect contact geometry. Gradients through contact events exhibited large discontinuities and high variance.
2) **Numerical instability**: Gradient magnitudes varied by orders of magnitude across timesteps, with contact/no-contact transitions producing gradient spikes that dominated the signal.
3) **Uninformative direction**: Even with gradient clipping and smoothing, the resulting morphology updates did not improve the objective—performance was statistically indistinguishable from random perturbations.

Implication: This explains why all existing RL-based co-design methods default to gradient-free outer loops. The contact-rich dynamics that make humanoid locomotion challenging for co-design also make analytical morphology gradients unreliable. Our finite-difference approach sidesteps this entirely by treating the simulator as a black box.

IV. Experimental Design

A. Primary Evaluation: Humanoid Locomotion

We focus our primary evaluation on full-body humanoid morphology optimization for several reasons:

1) **Complexity**: Contact-rich dynamics, 30+ DOF, and stability constraints represent the most challenging test case for co-design scalability.
2) **Motion quality**: Integration with Adversarial Motion Priors (AMP) [**?**] prevents morphology exploitation and ensures human-like motion transferable to hardware.
3) **Unexplored**: Full humanoid co-design with deep RL has not been demonstrated in any prior work. The closest system (RoboCraft [**?**]) operates on small soccer robots.
4) **Unexplored design space**: Joint oblique angles—rotational offsets in joint reference frames—represent a practical manufacturing parameter that affects kinematics and load distribution. Despite their practical relevance, they remain unexplored in prior co-design work.

1) Morphology Parameterization: We parameterize the humanoid with oblique angles $\phi \in \mathbb{R}^n$ representing joint orientation offsets from anatomical neutral, constrained to $\phi_i \in [-30, +30]$ for biomechanical plausibility. Our primary experiments use 6 parameters (hip and knee joints); we additionally evaluate scaling to higher dimensions.

2) Tasks:

- **Walking**: Steady-state bipedal locomotion at 1.0–1.5 m/s
- **Cartwheeling**: Dynamic full-body rotation with arm-ground contact
- **Jump Kick**: Explosive ballistic movement with aerial phase

3) AMP Integration: For humanoid locomotion, we use AMP [**?**] to provide style rewards from motion capture data:

$$r_{total}(s, a) = r_{task}(s, a) + \lambda \cdot \log D_\psi(s, a) \tag{10}$$

where D_ψ is a discriminator trained on human motion. This prevents unrealistic morphologies that exploit simulation artifacts while improving exploration through motion priors.

4) Finite-Difference Configuration: We use central finite differences with perturbation $\epsilon_{fd} = 1$ for oblique angles. Each perturbed design is evaluated over $N_{eval} = 50$ episodes to reduce stochastic variance. For $n = 6$ parameters, each gradient step requires $2 \times 6 + 1 = 13$ frozen-policy evaluations.

B. Generality Validation: Link Length Optimization

To validate framework generality beyond oblique angles, we conduct secondary experiments optimizing link lengths for humanoid walking. This demonstrates that PGHC is not tied to a specific morphology parameterization.

C. Evaluation Metrics

TABLE I
Task-Specific Evaluation Metrics

Task	Primary Metric	Secondary Metrics
Walking	Cost of Transport	Success rate, speed
Cartwheel	Success rate	Stability time, speed
Jump Kick	Peak height/distance	Success rate, energy

Cost of Transport (CoT) for locomotion tasks:

$$CoT = \frac{E_{total}}{m \cdot g \cdot d} \tag{11}$$

where E_{total} is energy consumed, m is robot mass, g is gravity, and d is distance traveled.

D. Baselines

1) **Fixed Morphology + RL**: Baseline with default morphology parameters
2) **CMA-ES Co-Design**: Evolutionary optimization with full policy retraining per candidate (matched parameter count)
3) **Random Morphology Sampling**: Random parameters within constraints
4) **BPTT Co-Design**: Differentiable physics gradient (included to document failure mode)
5) **Ablations**:
   - No stability gating (fixed iteration schedule)
   - No trust region (fixed step size)
   - Both disabled (naive FD gradient descent)

V. Results

A. Main Results: Oblique Angle Optimization

Table II shows PGHC achieves 12–18% improvements in task-specific metrics while maintaining high success rates and motion quality (AMP scores > 0.85).

TABLE II
PERFORMANCE ON HUMANOID MORPHOLOGY (OBLIQUE ANGLES)

Task	Baseline	PGHC	Improvement	Success
Walking (CoT)	0.45	0.38	15.6%	100% / 100%
Cartwheel (SR)	72%	89%	+17%	—
Jump Kick (Height)	1.2m	1.4m	+16.7%	87% / 94%

B. Sample Efficiency: Comparison to Evolutionary Methods

PGHC achieves comparable or superior performance to CMA-ES with dramatically fewer environment interactions. For the walking task, PGHC requires $\sim$30M environment steps (one policy training + FD evaluations) compared to $\sim$300M+ for CMA-ES (which retrains the policy for each of $\sim$100+ candidate designs). This $\sim 10\times$ reduction in sample complexity is a direct consequence of the Envelope Theorem: PGHC retrains the policy only when the stability gate re-triggers after a morphology update, while CMA-ES must retrain from scratch for every candidate.

C. BPTT Baseline Comparison

The BPTT baseline (differentiable physics through MJX) failed to produce meaningful morphology improvements: final CoT for walking was 0.44, statistically indistinguishable from the fixed-morphology baseline (0.45). This confirms the negative result in Section III and validates our finite-difference approach.

D. Ablation Studies

Both components are necessary: stability gating ensures the Envelope Theorem approximation is valid (policy is converged before FD evaluation), while trust regions prevent morphology updates that push the design outside the frozen policy's basin of attraction.

TABLE III
ABLATION STUDY: FRAMEWORK COMPONENTS

Configuration	Walking CoT	Training Time
PGHC (full)	0.38	12 hours
No stability gating	0.41 (7.9% worse)	18 hours
No trust region	0.40 (5.3% worse)	15 hours
Neither (naive FD-GD)	0.42 (10.5% worse)	20+ hours

E. Envelope Theorem Approximation Quality

We empirically validate the gradient approximation by comparing:

- **FD gradient (frozen policy)**: $\hat{\nabla}_\phi \mathcal{J}$ computed via finite differences with frozen θ^* (our method)
- **True gradient estimate**: Finite-difference with full policy retraining for each perturbed design (ground truth, but $\sim 100\times$ more expensive)

Cosine similarity between the frozen-policy FD gradient and the retrained FD gradient is 0.85–0.92 across design iterations, confirming that the Envelope Theorem approximation captures the dominant gradient direction. Critically, this similarity is highest in early iterations (when the policy is well-converged) and decreases only when cumulative morphology changes are large—exactly the regime where re-triggering the stability gate and retraining is appropriate.

F. Generality: Link Length Optimization

Table IV shows PGHC successfully optimizes link lengths for walking, achieving 8.9% CoT improvement. This validates framework applicability beyond oblique angles.

TABLE IV
GENERALITY DEMONSTRATION: LINK LENGTH OPTIMIZATION

Parameterization	DOF	CoT	Improvement
Baseline (fixed)	—	0.45	—
Link lengths	12	0.41	8.9%
Oblique angles	6	0.38	15.6%

G. Dimensionality Scaling

The framework scales to higher-dimensional design spaces. For oblique angle optimization on walking:

- 3 DOF (knees only): 6.2% CoT improvement
- 6 DOF (hips + knees): 15.6% CoT improvement
- 12 DOF (hips + knees + ankles): 18.1% CoT improvement

For higher dimensions, SPSA [**?**] can replace central FD to maintain constant per-step evaluation cost (2 evaluations regardless of dimensionality).

H. Critical Design Parameters

Analysis reveals hip and knee oblique angles account for the majority of walking CoT improvement, while upper body angles contribute primarily to cartwheel and jump kick performance—validating the biomechanical importance of lower-limb joint orientations for locomotion efficiency.

VI. Related Work

A. Evolutionary Co-Design

The dominant paradigm pairs evolutionary outer loops with RL inner loops. DERL [**?**] co-evolves morphology and control for modular robots. RoboCraft [**?**] uses CMA-ES for small soccer robots. TAME [**?**] applies genetic algorithms to task-agnostic morphology evolution. All require full policy retraining per candidate design. PGHC replaces the evolutionary outer loop with gradient descent, eliminating this cost.

B. Differentiable Co-Design

Ha et al. [**?**] compute first-order morphology gradients via implicit function theorem for trajectory-optimized quadrupeds. Desai et al. [**?**] use adjoint methods for similar systems. Both use trajectory optimization, not RL, and operate on simple robots. Transform2Act [**?**] learns to transform robot morphologies but does not optimize morphology parameters directly. Our BPTT negative result (Section III) explains why this approach has not been extended to RL-based co-design with contact.

C. Frozen-Controller Approaches

Strgar et al. [**?**] freeze a pretrained controller and evolve morphologies in a "zero-shot" setting—conceptually the closest to our approach. However, they use evolutionary search (not gradient descent) and voxel-based soft robots (not humanoids). PGHC provides the theoretical justification (Envelope Theorem) for why freezing works, and combines it with gradient descent for more efficient optimization.

VII. Discussion

A. Limitations

1) **Parameter count**: Our primary experiments optimize 6 morphology parameters. While we demonstrate scaling to 12 and show that SPSA enables further scaling, validation on hundreds of parameters remains future work. The algorithmic framework is parameter-count agnostic.
2) **FD noise**: Finite differences are susceptible to evaluation noise, requiring multiple rollout episodes per perturbation. This cost scales linearly with required precision.
3) **Sim-to-real transfer**: Optimized morphologies are validated in simulation. Real-world deployment requires domain randomization and robustness analysis.
4) **Local optima**: Like all gradient-based methods, PGHC can converge to local optima. Hybrid evolutionary-gradient approaches or random restarts may improve global search.
5) **Manufacturing constraints**: Optimized parameters may not correspond to easily manufacturable configurations without additional constraints.

B. Why Not Just Use CMA-ES?

A natural question is whether the complexity of PGHC is justified over simpler evolutionary baselines. The key advantages are:

1) **No retraining per candidate**: CMA-ES evaluates each candidate by training a full RL policy ($\sim$hours). PGHC evaluates by rolling out a frozen policy ($\sim$seconds).
2) **Gradient information**: First-order methods exploit local curvature, enabling faster convergence in smooth regions of the design landscape.
3) **Theoretical grounding**: The Envelope Theorem provides formal justification for why frozen-policy evaluation is valid, rather than relying on empirical observation.

CMA-ES may still be preferable for highly non-smooth or multi-modal design landscapes where gradient information is less useful.

C. Future Work

1) **Higher-dimensional validation**: SPSA-based scaling to 50+ morphology parameters
2) **Hardware validation**: Real-world testing of optimized morphologies
3) **Multi-objective optimization**: Balance efficiency, robustness, and cost
4) **Expanded design spaces**: Joint optimization of geometry, actuators, and materials
5) **Hybrid evolutionary-gradient methods**: CMA-ES for global search + PGHC for local refinement

VIII. Conclusion

We present PGHC, a scalable framework for robot co-design that introduces three elements novel to the field: (1) the Envelope Theorem as theoretical justification for frozen-policy morphology optimization, (2) finite-difference gradient descent replacing population-based search for the morphology outer loop, and (3) dynamic stability gating and adaptive trust regions that enforce the conditions under which gradient decoupling remains valid. We demonstrate PGHC on full-body humanoid locomotion—the first such demonstration with deep RL—achieving 12–18% performance improvements across diverse tasks. We additionally document why differentiable physics fails for morphology optimization through contact, explaining the field's reliance on gradient-free methods and motivating our finite-difference approach. PGHC establishes that principled gradient-based co-design is both theoretically justified and practically effective, opening a new direction between the evolutionary and differentiable physics paradigms.

References